Zhihan Lei

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in **Machine Learning** (GPA: 3.7) — *Dec 2026*

Emory University — **Atlanta, GA**
Bachelor of Science in **Physics**, *Minor* in **Computer Science** (GPA: 3.8) — *Dec 2023*

PUBLICATIONS & MANUSCRIPTS

Published:

Wang, B., **Lei, Z.**, et al. *SpecDrop: Parameter-Free Category-Conditioned Routing for Modular Specialization*. NeurIPS 2026, *second author*.

Hu, Y., **Lei, Z.**, et al. *GRAG: Graph Retrieval-Augmented Generation*. Findings of NAACL 2025, *second author; 395 citations*.

Hu, Y., **Lei, Z.**, et al. *CG-RAG: Research Question Answering by Citation Graph Retrieval-Augmented LLMs*. SIGIR 2025, *second author; 27 citations*.

Preprints:

Lei, Z., et al. *Learning Reusable Agent Subroutines from Experience*. Under review at ICLR 2027, *first author*; arXiv, code.

Kumar, A., **Lei, Z.**, et al. *Learning to Construct Practical Agentic Systems*. arXiv:2606.00189.

RESEARCH EXPERIENCE

Carnegie Mellon University — LLM Agents & Reasoning — **Pittsburgh, PA**
Graduate Researcher (Advisor: Prof. William Cohen) — *Jan 2026 - Present*

- Proposed **ARISE**, a **gradient-free self-improvement** method: an LLM agent mines its own successful traces into a library of **natural-language subroutines**, each executed by a separate LLM call.
- Across 5 tasks (MuSR, RuleArena, NatPlan) on **DeepSeek-V3** and **Gemini 2.5 Flash**, beat the strongest of 6 baselines (incl. AWM, ACE, ASI) by **+6.1–10.8pp** on average, and the source ReAct agent by up to **+45.4pp** (NBA 28.8%→74.2%).
- **Execution-interface ablation**: holding the induced text fixed, serving it as skill docs or Python tools scores **7.9–19.1pp lower**; libraries also **transfer unchanged across models** (+10.7–32.8pp on all 5 tasks).
- Co-authored a companion **cross-benchmark study** mapping when induction helps across **19 reasoning tasks**: gains scale with baseline-tool weakness, e.g. **+8.2pp on MedCalc at ∼60% fewer input tokens**.

Emory Machine Learning Lab — Graph Learning & Retrieval — **Atlanta, GA**
Researcher (Advisor: Prof. Liang Zhao) | **GitHub** — *Apr 2024 - Feb 2025*

- **GRAG** (NAACL 2025 Findings): divide-and-conquer subgraph retrieval at linear time with soft pruning, dual text/graph-view prompting; beat prior SOTA graph retriever G-Retriever on WebQSP multi-hop QA.
- **CG-RAG** (SIGIR 2025): chunk-level citation graph + Lexical-Semantic Graph Retrieval entangling sparse/dense via message passing; generalizes existing hybrid and graph retrievers.

PROFESSIONAL EXPERIENCE

Amazon — **Seattle, WA**
Software Engineer Intern (return offer) — *May 2026 - Aug 2026*

- Built an **agentic analytics compiler** replacing manual Redshift→PartiQL migration: an LLM agent transpiles queries, **validates semantic correctness by executing against benchmark cases**, and auto-generates dashboards from context hints.
- **Designed and owned the service end-to-end** — Figma prototype through backend APIs, permissions, CDK infrastructure, and integration tests (∼15k-LOC service; TypeScript/Java/Python); **shipped to internal beta**.

ByteDance — **Beijing, China**
Machine Learning Engineer Intern — *Jul 2023 - Sep 2023*

- Fine-tuned face-matching model for account security: **62% → 83% live precision** (human-reviewed); deployed as real-time identity verification.

PROJECTS

SpecDrop: Modular Specialization | CMU | NeurIPS 2026 — *Feb 2026 - May 2026*

- **Parameter-free category-conditioned routing**: repurposes dropout to drive modular specialization with no learned gating network, with **stability-guided freezing** to cut training compute; designed the **specialization metrics** (module–category MI, pruning sensitivity) and per-domain eval on CIFAR-100 ResNets and **60–125M MoE LMs**.

SKILLS

Research: LLM Agents & Reasoning, Test-Time Compute, Self-Improving Agents, Tool Synthesis & Skill Discovery, LLM Evaluation, Modular Specialization (MoE), Retrieval-Augmented Generation, Graph Neural Networks
Methods: Deep Learning, Transformer/Attention, NLP, Fine-tuning, Probabilistic Graphical Models, Bayesian Inference
Languages & Tools: Python (PyTorch, Hugging Face, NumPy, Pandas), SQL; CUDA, AWS, Spark, Docker, Git, Linux